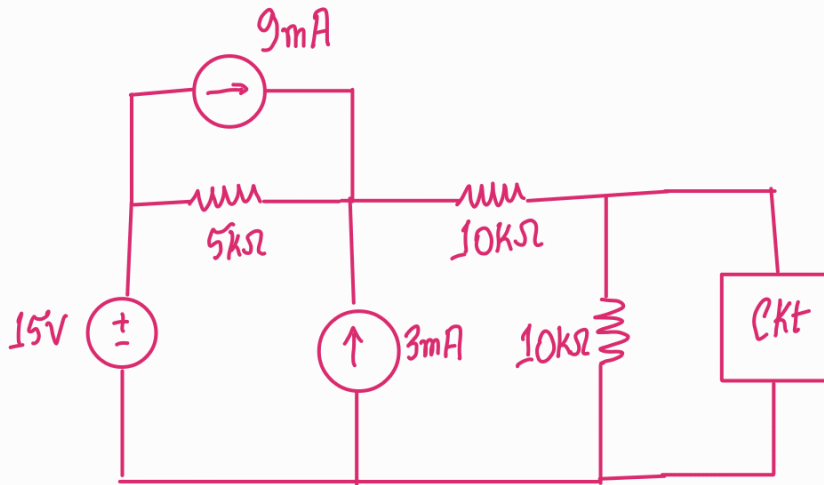


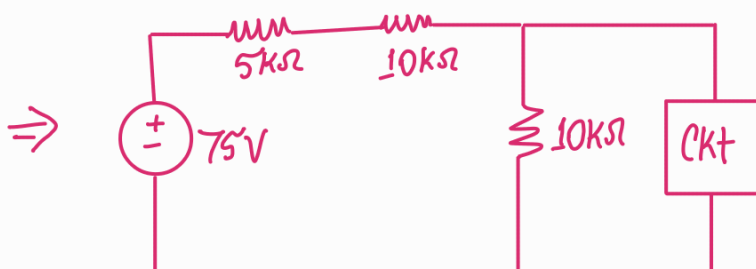
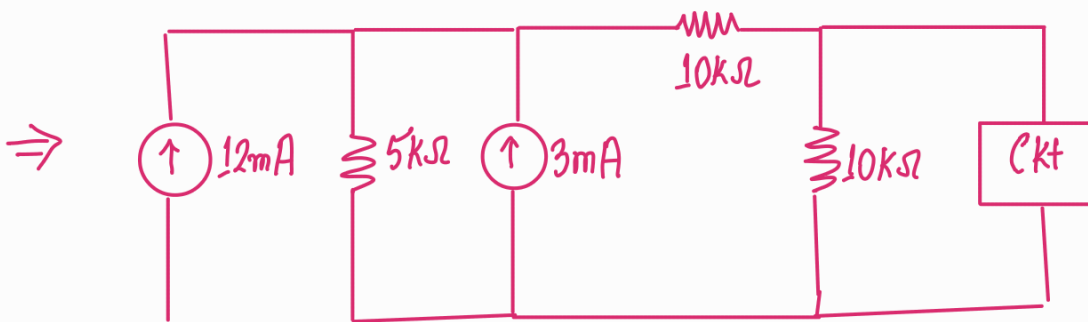
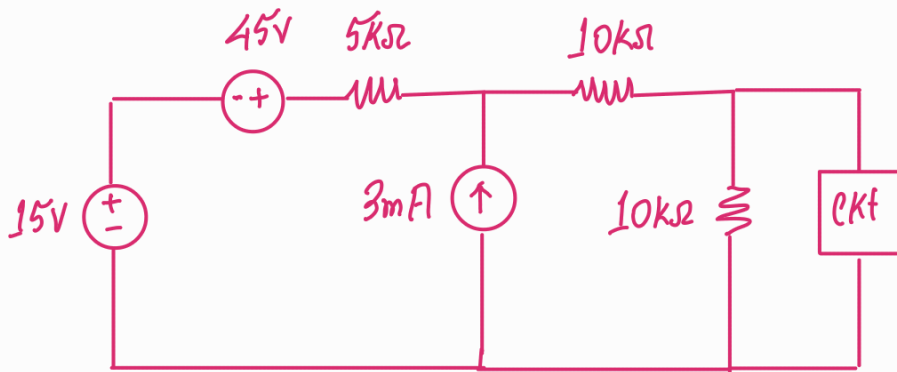
Set A

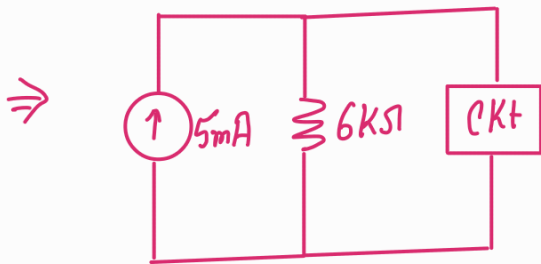
Q1

Ⓐ Left hand side of the ckt —

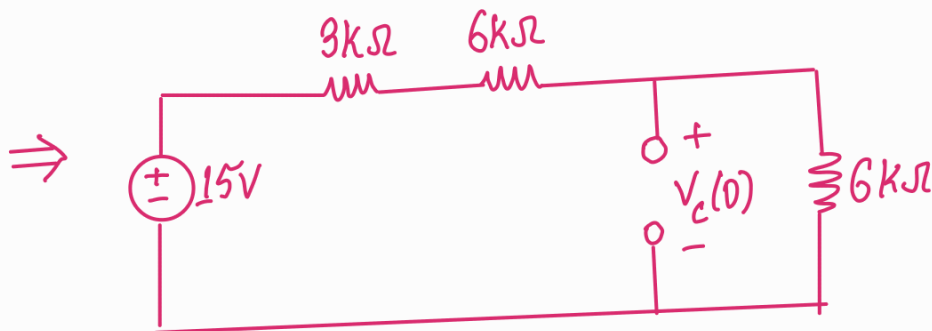
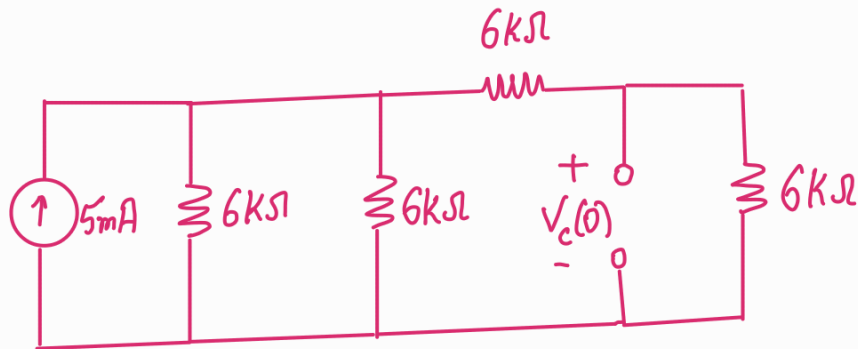


Simplifying —





now $t=0^-$ ckt at capacitor steady state —

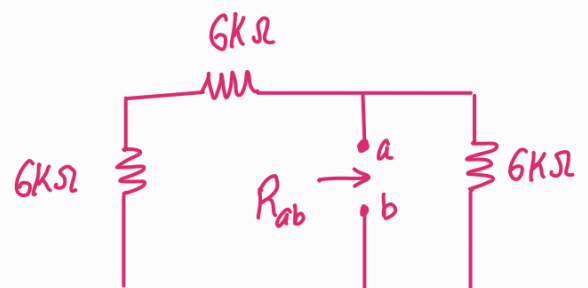
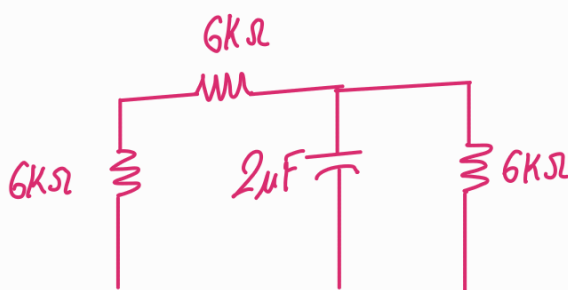


$$\therefore V_c(0) = \frac{6k}{3k+6k+6k} \times 15 \text{ [voltage divider]}$$

$$= 6V$$

Since the $t>0$ ckt doesn't have any srcs, $V_c(\infty) = 0V$

(b) $t>0$ ckt —



disconnecting load to calculate R_{Th} (R_{ab} in diagram)

$$R_{Th} = R_{ab} = 6k \parallel (6k + 6k) = 4k\Omega$$

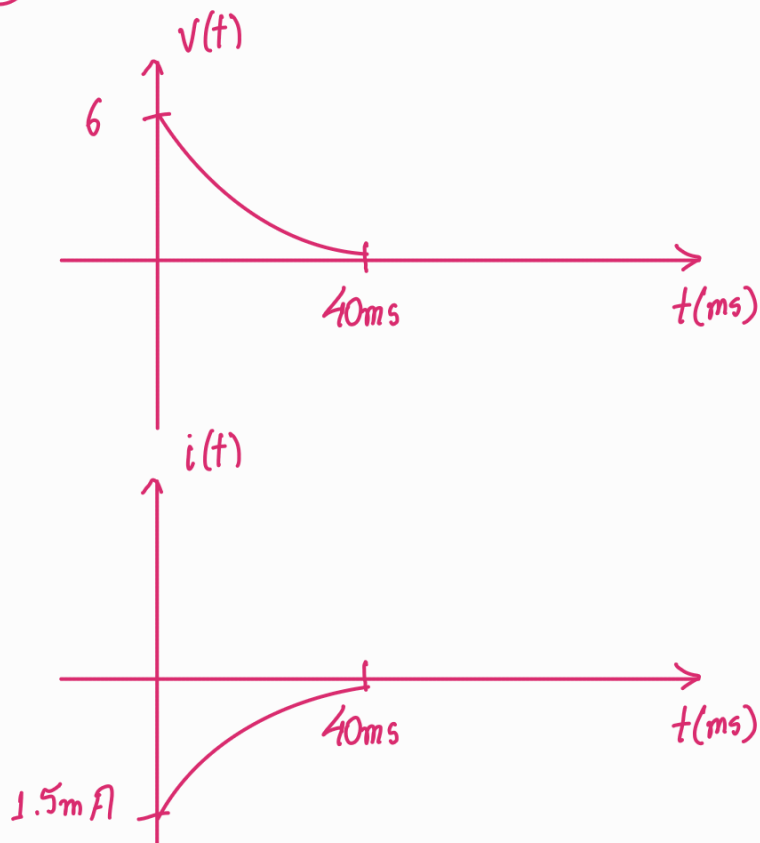
$$\therefore \tau = R_{Th}C = 4k \times 2\mu = 8ms$$

③ We know, for $t > 0$:

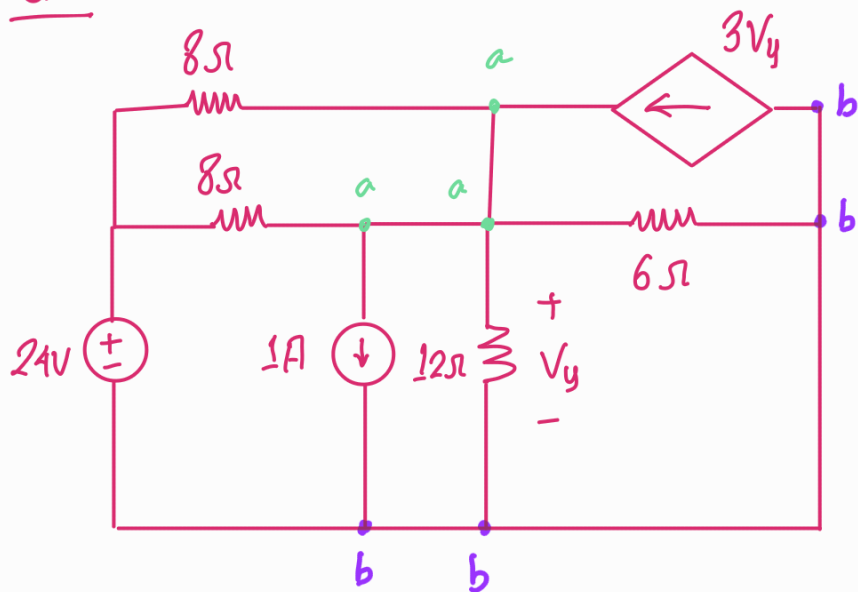
$$\begin{aligned} V_c(t) &= V_c(\infty) + [V_c(0) - V_c(\infty)]e^{-t/\tau} \\ &= 6e^{-t/8 \times 10^{-3}} \\ &= 6e^{-125t} \text{ V} \end{aligned}$$

$$\begin{aligned} \text{④ } i(t) &= i_c(t) = C \frac{dV_c(t)}{dt} \\ &= 2 \times 10^{-6} \times 6e^{-125t} \times (-125) \\ &= -1.5e^{-125t} \text{ mA} \end{aligned}$$

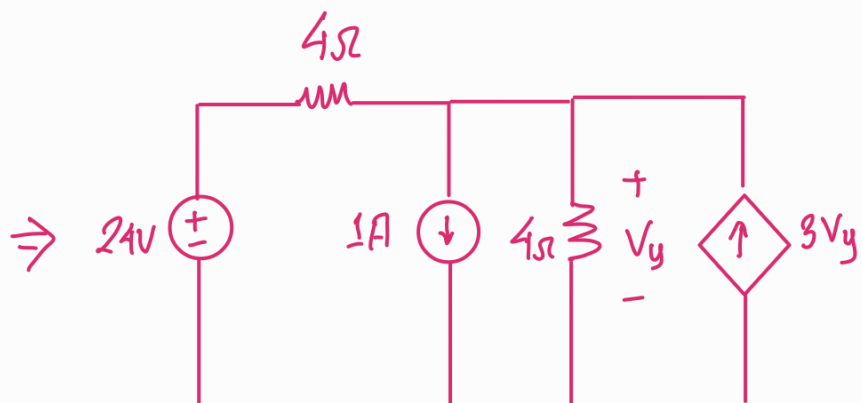
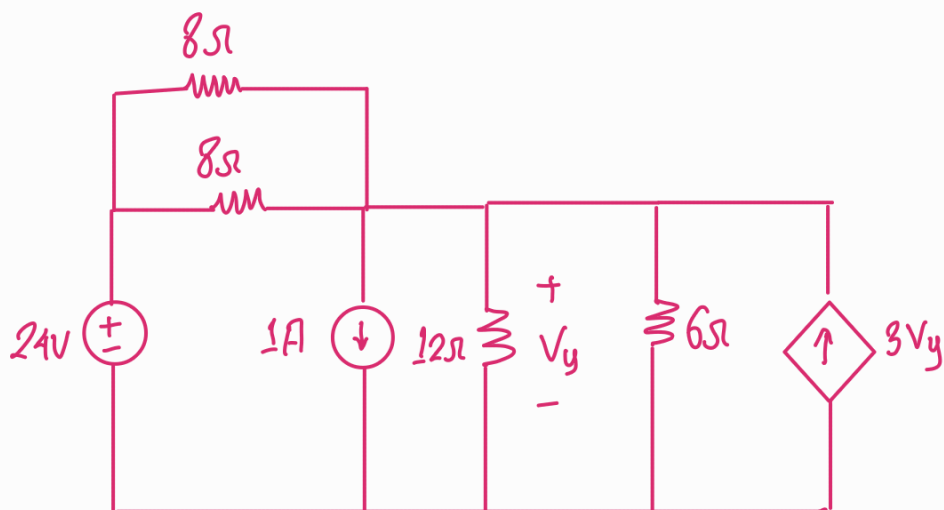
⑤



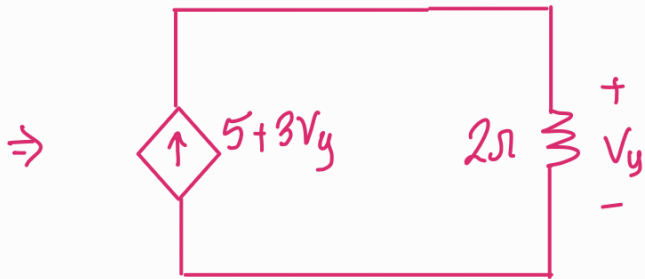
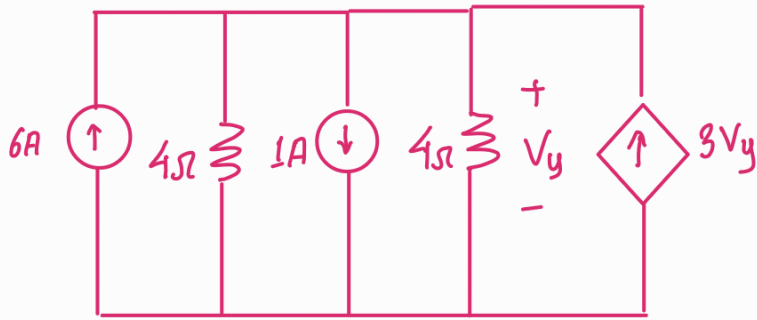
Q2



Redrawing—



Solving by Src. transformation —

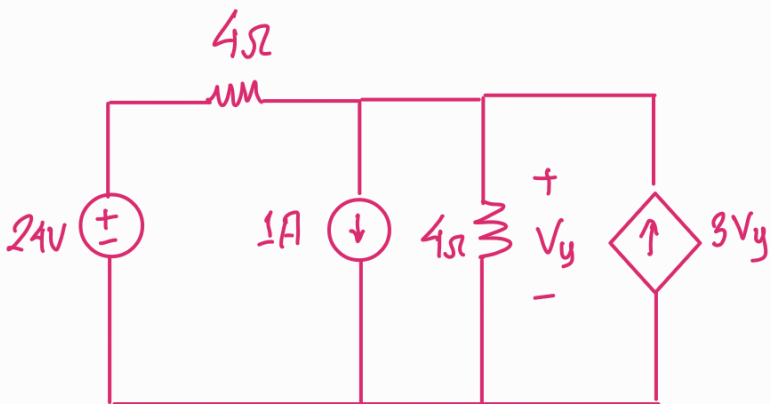


$$\therefore V_y = 2 \times (5 + 3V_y)$$

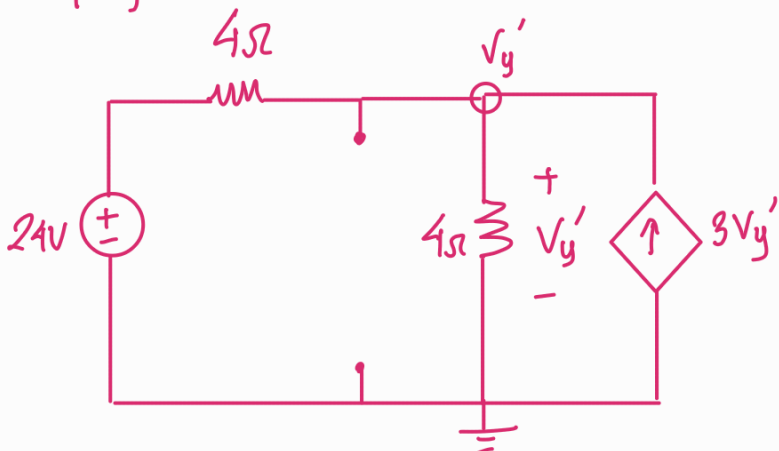
$$\Rightarrow V_y = 10 + 6V_y$$

$$\Rightarrow -5V_y = 10 \Rightarrow V_y = -2V$$

Solving by superposition —



Keeping the 24V src ON —



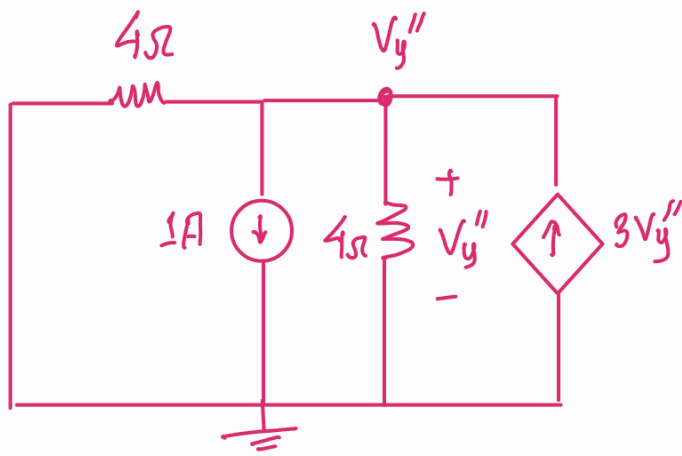
KCL @ node y' —

$$\frac{V_y' - 24}{4} + \frac{V_y'}{4} - 3V_y' = 0$$

$$\Rightarrow V_y' - 24 + V_y' - 12V_y' = 0$$

$$\Rightarrow V_y' = -2.4V$$

Keeping the 1A src ON—



KCL @ node y'' —

$$\frac{V_y''}{4} + 1 + \frac{V_y''}{4} - 3V_y'' = 0$$

$$\Rightarrow V_y'' + 2 - 6V_y'' = 0$$

$$\Rightarrow V_y'' = 0.4V$$

$$\therefore V_y = V_y' + V_y'' = -2.4 + 0.4 = -2V$$

Q3

According to the Question —

V	I
2	3
-6	-1

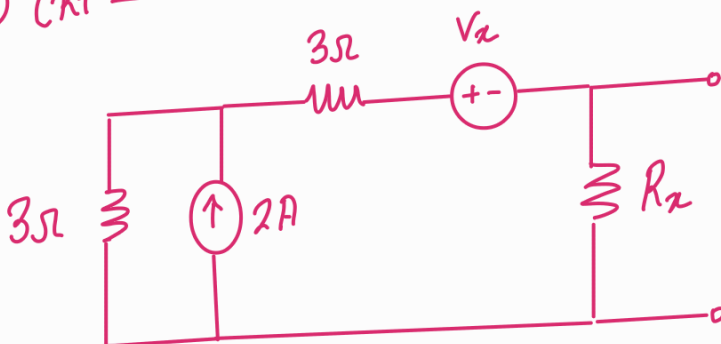
$$IV \rightarrow \frac{-6-2}{V-2} = \frac{-1-3}{I-3}$$

$$\Rightarrow 2I - 6 = V - 2$$

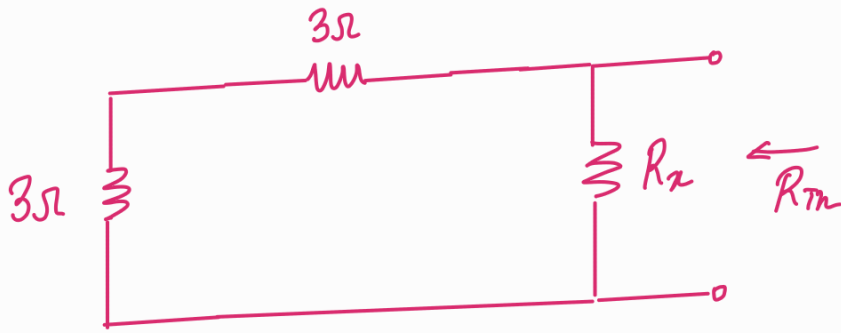
$$\Rightarrow I = \frac{V}{2} + 2$$

Comparing to $I = \frac{V}{R_{Th}} - \frac{V_{Th}}{R_{Th}}$, $R_{Th} = 2\Omega$ & $V_{Th} = -4V$

(a) Ckt —



Turning off all the srcs to find R_{Th} —



$$R_{Th} = R_x \parallel (3+3)$$

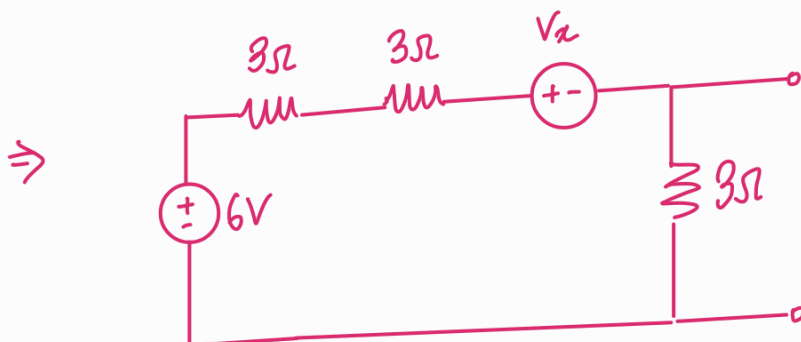
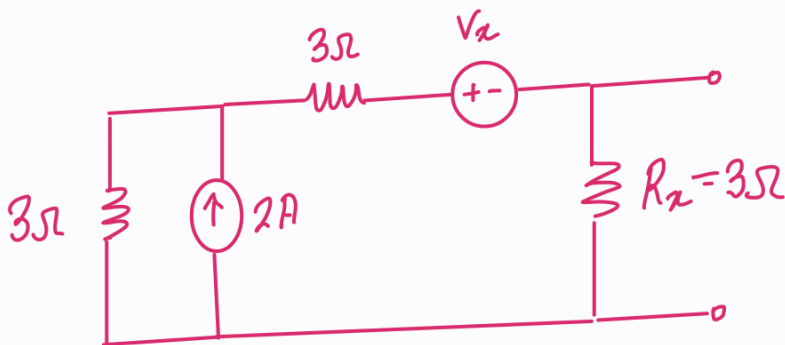
$$\Rightarrow 2 = \left(\frac{1}{R_x} + \frac{1}{6} \right)^{-1}$$

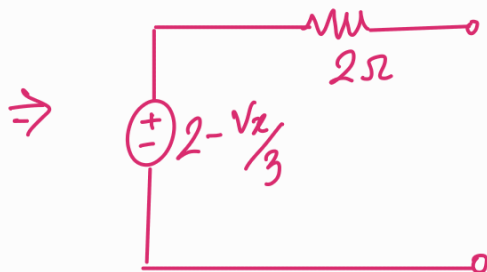
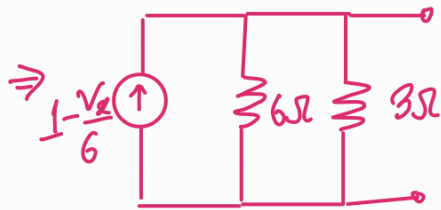
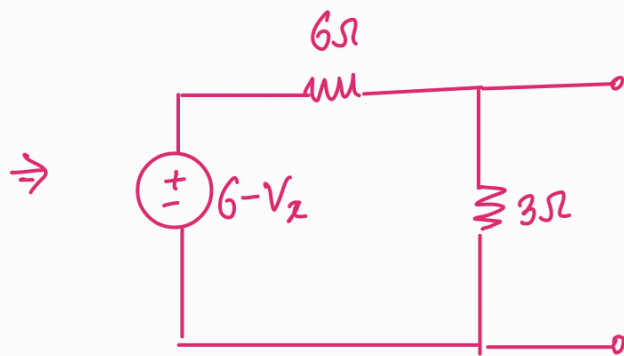
↑
found from IV

$$\Rightarrow \frac{1}{2} = \frac{1}{R_x} + \frac{1}{6}$$

$$\Rightarrow \frac{1}{R_x} = \frac{3-1}{6} \Rightarrow R_x = 3\Omega$$

⑥ Simplifying the ckt to match Thevenin's ckt —





$$\therefore 2 - \frac{V_2}{3} = V_{Th} \Rightarrow 2 - \frac{V_2}{3} = -4$$

$\uparrow$
found from IV

$$\Rightarrow V_2 = 18V$$

Q4

① V_1 leads V_2 by $\Delta t = 6ms$

$$T = 16ms$$

$$\therefore \Delta\phi = \frac{6}{16} \times 360^\circ = 135^\circ$$

$$\textcircled{b} \quad V_1(t) = 8 \sin \left(2\pi \times \frac{1}{16 \times 10^{-3}} \times t + \frac{2}{16} \times 360^\circ \right) \text{ V}$$

$$= 8 \sin (125\pi t + 45^\circ) \text{ V}$$

$$V_2(t) = 6 \sin (125\pi t - \frac{4}{16} \times 360^\circ) \text{ V}$$

$$= 6 \sin (125\pi t - 90^\circ) \text{ V}$$

$$\phi_{V_1} - \phi_{V_2} = 45 - (-90) = 135^\circ \text{ [verified]}$$

for cosine —

$$V_1(t) = 8 \cos (125\pi t - \frac{2}{16} \times 360^\circ) \text{ V}$$

$$= 8 \cos (125\pi t - 45^\circ) \text{ V}$$

$$V_2(t) = 6 \cos (125\pi t - \frac{8}{16} \times 360^\circ) \text{ V}$$

$$= 6 \cos (125\pi t - 180^\circ) \text{ V}$$

$$\phi_{V_1} - \phi_{V_2} = -45 - (-180) = 135^\circ \text{ [verified]}$$